# Harshal Parekh

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Surat, India | Open to relocation | Available to join immediately

## Summary

2026 Computer Engineering graduate with 8 months of industry experience as an AI Engineer. Built and deployed diffusion-based image & video generation pipelines for 5+ brands using FastAPI and Docker. Also shipped a production multi-branch ERP (.NET, React, PostgreSQL) with the help of AI coding agents. Seeking Software, AI/ML, and Backend roles.

## Experience

**Junior AI Engineer** Nov 2025 – Apr 2026
*NexAI Labs* *Surat, India*
- Built diffusion-model image generation pipelines for 5+ garment and jewellery brands, reducing manual design effort by 60%.
- Automated end-to-end image & content generation using n8n multi-agent workflows that unified multiple model APIs.
- Extended the system to AI video generation, improving client content delivery speed by 40%.
- Deployed AI services as REST APIs (FastAPI + Docker) on AWS EC2 with media storage on S3 and Cloudflare R2.

**AI Engineer Intern** Aug 2025 – Oct 2025
*NexAI Labs* *Surat, India*
- Developed prompt workflows and API integrations for AI automation pipelines with Pandas/NumPy processing.
- Deployed and tested FastAPI-based ML applications on Google Cloud.

## Projects

**Multi-Branch Gaming Cafe ERP** | *C#, .NET 8, React, PostgreSQL, SignalR, Redis, Docker* GitHub
- Built and deployed a full-stack ERP for multi-branch gaming cafés managing 100+ PCs with real-time session & billing sync.
- Developed ASP.NET Core API (Clean Architecture, 30+ controllers, EF Core) + 16-module React frontend with SignalR hubs and a Windows agent for remote PC control.
- Implemented usage-based billing, split payments, JWT role-based auth, and deployed 7 Docker services on Oracle Cloud with Nginx + SSL.
- Built using AI coding agents (Claude Code, Cursor, Antigravity) – wrote specs, scoped tasks, and reviewed/debugged their code.

**Prism – AI Research Learning Studio** | *Python, FastAPI, Gemini, SQLite, JavaScript* Live | GitHub
- Built a multi-user PDF research assistant with authentication, multi-turn conversations, and paper comparison across up to 5 documents.
- Grounded Gemini responses using SQLite FTS5/BM25 retrieval; added interactive notebooks, quizzes, diagrams, and numerical lessons.
- Deployed on Oracle Cloud with Nginx/HTTPS and GitHub Actions CI/CD (38 tests, automated deploy + health-check rollback).

**AI CCTV Safety Surveillance** | *Python, YOLOv8, PyTorch, OpenCV, Streamlit* GitHub
- Fine-tuned YOLOv8 on a 10-class PPE dataset to detect missing hard hats, masks, and vests in images and live video.
- Logged violations with timestamps and cropped snapshots; built Streamlit app with live WebRTC, analytics, and PDF/CSV reports.

## Technical Skills

**Languages:** Python, SQL, C#, Java (basic)
**AI / ML:** Diffusion Models, YOLOv8, PyTorch, OpenCV, LangChain, RAG, Qdrant, Gemini API, LLMs, Prompt Engineering, Hugging Face
**Backend & APIs:** FastAPI, Flask, ASP.NET Core, EF Core, SignalR, REST APIs, JWT, Pydantic, Swagger
**DevOps & Cloud:** Docker, Nginx, Linux, GitHub Actions, AWS (EC2, S3), Google Cloud, Oracle Cloud, Cloudflare R2, CI/CD
**Databases:** PostgreSQL, SQLite, SQL, Redis
**Frontend & Tools:** React, Tailwind CSS, Streamlit, n8n, Claude Code, Cursor, Antigravity

## Education

**C.K. Pithawala College of Engineering and Technology** Surat, India
*B.E. in Computer Engineering – CGPA: 8.41 / 10* *Aug 2022 – May 2026*

## Certifications

**Ethical Hacking & Penetration Testing** – C-DAC Noida (training + virtual internship)
**AWS Cloud Practitioner (CLF-C02) Exam Prep** – Udemy
**Python, Data Analysis & AI Foundation** – SAP Code Unnati (Edunet Foundation)
**Google Cloud Skills Boost (22 badges)** – Gen AI with Gemini, Cloud Run, Vision API, Prompt Design (Profile)